

Co-Design and Evolutionary Optimization of Periodic Event-Triggered Robust Control for LPV Systems^{*}

Lucas A. L. Oliveira^{*,**} Miguel L. Rodrigues^{***}
Kevin Guelton^{**} Koffi M. D. Motchon^{**}
Valter J. S. Leite^{*,***}

^{*} Graduate Program on Mathematical and Computational Modeling (PPGMMC) of the CEFET-MG, Belo Horizonte, 30.510-000, MG, Brazil (e-mail: arantesl@hotmail.com).

^{**} CReSTIC UR3804, Université de Reims Champagne-Ardenne, 51100 Reims, France (e-mail: kevin.guelton,koffi.motchon@univ-reims.fr)

^{***} Department of Mechatronics Engineering, CEFET-MG - Campus Divinópolis, Divinópolis, 35503-822, MG, Brazil, PPGEL, and PPGMMC (e-mail: valter@ieee.org, miguellukas52@gmail.com).

Abstract:

Networked control systems (NCSs) play an important role in Industry 4.0 and the Internet of Things (IoT), offering benefits such as operational cost reduction, energy efficiency, and architectural flexibility. However, communication constraints, including limited bandwidth and delays, present challenges to closed-loop stability and performance. Event-triggered control (ETC) has emerged as a solution to optimize network usage while ensuring stability. This work focuses on the co-design of periodic event-triggered robust controllers for linear parameter-varying (LPV) systems to ensure the stabilization of the closed-loop system. Utilizing parameter-dependent looped-functionals, second-order polynomial negativity, and Pólya's theorem, LMI-based stability conditions are developed under a static event-triggered mechanism (ETM) structure. A differential evolution algorithm is introduced to reduce control signal updates compared to convex optimization procedures. Numerical validation demonstrates the effectiveness of the proposed approach, highlighting its advantages over existing methods.

Keywords: Periodic event-triggered control, LPV systems, looped-functional, robust control, differential evolution algorithm.

1. INTRODUCTION

Networked control systems (NCSs) are widely used in industrial systems due to Industry 4.0 and the Internet of Things (IoT). Such a framework offers several advantages, including lower operational costs, reduced energy consumption, and architectural flexibility since the control elements can be distributed over digital communication networks, including wireless ones (Hespanha et al., 2007; Zhang et al., 2020). However, constraints from limited communication resources, including restricted bandwidth and limited energy availability, along with challenges inherent to communication networks, like transmission delays, quantization, and packet dropouts, can compromise the closed-loop stability of the system or lead to significant performance degradation (Peng and Li, 2018).

To ensure closed-loop stability and performance while optimizing network resource utilization, event-triggered control (ETC) has emerged as an effective strategy for balancing network usage and stability criteria (Tabuada,

2007; Hespanha et al., 2007). The ETC operates by transmitting information over the network and updating the control signal only when specific criteria, such as state or output conditions, are satisfied. Such a mechanism prevents unnecessary data transmissions and releases network communications resources for other devices or processes belonging to the NCS. The design of event-triggered mechanisms (ETMs) can follow two paths: emulation (Girard, 2015; Zhang et al., 2020; Xu et al., 2024) and co-design (Coutinho and Palhares, 2021; de Souza et al., 2022; Peixoto et al., 2024). In emulation, an ETM is designed for an available stabilizing controller. In co-design, on the other hand, both the stabilizing controller and the ETM are designed simultaneously, optimizing the network resources used with guaranteed stability.

A critical aspect of ETM design is selecting the triggering function. Literature categorizes strategies into static (Tabuada, 2007; de Souza et al., 2022; Xu et al., 2024) and dynamic (Girard, 2015; Coutinho and Palhares, 2021; Rouamel et al., 2023; de Souza et al., 2024), all aiming to minimize network transmissions. In general, the design of ETMs parameters finds convex formulations in the literature (Zhang et al., 2020; de Souza et al., 2024). However,

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these formulations are indirect optimization procedures and, consequently, do not achieve an optimum tuning of ETM. This limitation has led to exploring alternative approaches, like combining differential evolution (DE) algorithms (Storn and Price, 1997; Gonçalves et al., 2024) with convex methods, to directly optimize ETM performance (Rodrigues et al., 2024).

System measurements in ETCs can be performed in continuous (Tabuada, 2007) or discrete-time (de Souza et al., 2022), according to the plant temporal domain. However, with the advent of sampled-data systems (Fridman, 2010; Seuret, 2012), a mixed framework arises, where the plant evolves in continuous-time domain while the periodic ETM and the controller are designed in discrete-time one. Such an alternative is well-suited for digital devices operating through communication networks. Note that plant sensing can be performed either periodically (Heemels et al., 2013) (commonly referred to as Periodic Event-Triggered Control (PETC)) or aperiodically (Xu et al., 2024). In this context, the input-delay approach (Fridman, 2010) can be employed to model the sampling phenomenon, enabling the use of looped Lyapunov-Krasovskii functionals (LKF) (Seuret, 2012) to derive stability conditions in terms of Linear Matrix Inequalities (LMIs). A notable approach to reduce LKF conservatism involves constructing them to produce a time-derivative that depends polynomially on the delay parameter. Negativity lemmas have been proposed to ensure the negativeness of LKFs time-derivative, but the process becomes challenging whenever parameter-dependent systems are involved. Recently, Oliveira et al. (2025) proposed an effective solution to this challenge by means of Pólya's Theorem.

Despite the extensive literature on ETM and controller co-design and stability conditions for PETCs, there are unsolved issues, particularly concerning sampled-data control of linear parameter-varying (LPV) systems. In Peixoto et al. (2024), co-design conditions for parameter-dependent controllers and ETMs in LPV systems were proposed under denial-of-service and stochastic deception attacks. This approach assumes that the time-varying parameters of the controller evolve continuously, while only the system states are periodically sampled and transmitted based on event-triggered mechanisms. However, this assumption may be unrealistic, requiring the transmission of state and parameters in real-world applications. In Zhang et al. (2020), the parameter mismatch between the plant and sampled controller is addressed for PETC emulation design in LPV systems. Moreover, the proposed LMI conditions are limited to the emulation approach, relying on a pre-designed controller and not accounting for the co-design of the controller and ETM.

This work focuses on the co-design of PETC with robust controllers for LPV systems, where the plant states and parameters evolve in continuous-time, and the time-varying parameters are unavailable for feedback. The proposed approach employs parameter-dependent looped-functionals, whose derivatives yield second-order polynomials, with negativity ensured via Pólya's theorem. LMI-based conditions are derived to guarantee the asymptotic stability of the closed-loop LPV system under a static ETM that regulates state transmissions. A two-stage optimization procedure using DE algorithms is introduced to minimize

control signal updates compared to standard periodic sampling. The proposed methodologies are validated through a numerical example, showcasing improved performance relative to existing approaches in the literature.

Notations: The sets of real and natural numbers are denoted by $\mathbb{R}$ and $\mathbb{N}$, respectively. $W \in \mathbb{R}^{n \times m}$ and $x \in \mathbb{R}^n$ represent an $n \times m$ matrix with real entries and a vector with n components, respectively. $W \in \mathbb{S}^n$ denotes symmetric matrices of size $n \times n$ with real entries. The transpose of W is denoted by $W^\top$. $\mathbb{K}$ represents the set of vector-valued functions in $\mathbb{R}^n$ that are differentiable in $t \in [0, T)$. The Euclidean norm of $W \in \mathbb{R}^n$ is represented by $\|W\|$. $\text{diag}\{W_1, W_2\}$ is a block diagonal matrix composed of blocks W_1 and W_2 . $\text{He}\{W\}$ represents $W + W^\top$. In square matrices, the symbol $\star$ denotes symmetric transposed blocks. The matrices $\mathbf{I}$ and $\mathbf{0}$ stand for the identity and zero matrices of appropriate dimensions, respectively.

2. PROBLEM FORMULATION

Consider the following continuous-time LPV system:

$$\dot{x}(t) = A(\alpha(t))x(t) + B(\alpha(t))u(t) \quad (1)$$

where $x(t) \in \mathbb{R}^n$ is the state vector and $u(t) \in \mathbb{R}^{n_u}$ is the control input vector. The matrices $A(\alpha(t))$ and $B(\alpha(t))$ depend on time-varying parameters $\alpha(t) = [\alpha_1(t), \dots, \alpha_N(t)]^\top \in \mathbb{R}^N$ and belong to a polytope with N known vertices with $[A(\alpha(t)) \ B(\alpha(t))] = \sum_{i=1}^N \alpha_i(t) [A_i \ B_i]$, $\alpha(t) \in \Lambda_N$, where Λ_N is the unit simplex given by $\Lambda_N = \{\alpha \in \mathbb{R}^N : \sum_{i=1}^N \alpha_i = 1, \alpha_i \geq 0, i = 1, \dots, N\}$. We assume that the derivatives of $\alpha(t)$ are limited to known values. Also, any matrix $Y(\alpha(t)) = \sum_{i=1}^N \alpha_i(t) Y_i$ has its time derivative denoted by $\dot{Y}(\alpha(t)) = \sum_{i=1}^N \dot{\alpha}_i(t) Y_i$.

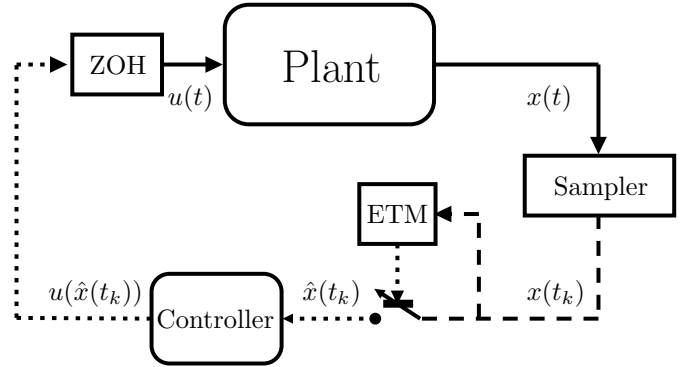


Fig. 1. Topology of the event-triggered control.

The control topology shown in Fig. 1 is addressed in this paper. The plant states $x(t)$, are measured at periodic sampling instants t_k , $t_k = kT$, with $k \in \mathbb{N}$ and $T > 0$. At each sampling instant, the state is transmitted to the controller only when the ETM generates an event. Then, the control signal is updated and kept constant by a zero-order holder (ZOH) until the next event. Because of periodic samplings, the Zeno behavior does not show up.

Assuming that the time-varying parameters are not available online for feedback, a robust control law based on the last transmitted sample, $\hat{x}(t_k)$, is considered:

$$u(t) = K\hat{x}(t_k), \forall t \in [t_k, t_{k+1}), \quad (2)$$

where $K \in \mathbb{R}^{n_u \times n}$ is the robust gain to be designed. The last transmitted state $\hat{x}(t_k)$ kept constant by the ZOH is defined as:

$$\hat{x}(t_k) = \begin{cases} x(t_k), & \text{if an event occurs,} \\ \hat{x}(t_{k-1}), & \text{otherwise.} \end{cases} \quad (3)$$

From (1) and (2), the closed-loop dynamics is given by:

$$\dot{x}(t) = A(\alpha(t))x(t) + B(\alpha(t))K\hat{x}(t_k), \quad \forall t \in [t_k, t_{k+1}). \quad (4)$$

Problem 1. Given the continuous-time LPV system (1) and the robust state-feedback control law (2), design the robust gain K and an event-trigger control strategy such that the closed-loop system (4) is asymptotically stable, reducing the number of control signal updates in comparison to the synchronous periodic sampling measures.

2.1 ETC strategy

We define the triggering condition $\ell : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ as:

$$\ell(x(t_k), \delta(t_k)) = \delta^\top(t_k) \mathcal{Q}_\delta \delta(t_k) - x^\top(t_k) \mathcal{Q}_x x(t_k) > 0, \quad (5)$$

where $\mathbf{0} < \mathcal{Q}_\delta \in \mathbb{S}^n$, $\mathbf{0} < \mathcal{Q}_x \in \mathbb{S}^n$, and $\delta(t_k)$ is the difference between the last transmitted state and the current state sampled t_k , that is $\delta(t_k) = \hat{x}(t_k) - x(t_k)$, and (4) can be rewritten, $\forall t \in [t_k, t_{k+1})$, as:

$$\dot{x}(t) = A(\alpha(t))x(t) + B(\alpha(t))K(x(t_k) + \delta(t_k)). \quad (6)$$

Therefore, when $\ell(x(t_k), \delta(t_k)) > 0$, an event occurs and $\hat{x}(t_k)$ is updated as in (3).

3. AUXILIARY RESULTS

We use the input-delay approach proposed by Fridman (2010) to rewrite the system (6) as:

$$\dot{x}(t) = A(\alpha(t))x(t) + B(\alpha(t))K(x(t-\tau(t)) + \delta(t-\tau(t))), \quad (7)$$

where $\tau(t)$ is a piecewise linear time-varying delay function defined as $t - t_k = \tau(t) \in [0, T]$, with $\dot{\tau}(t) = 1$.

To ensure the stability of the closed-loop system (6) considering the event-triggering rule, we use the following lemma based on the looped-functional approach:

Lemma 2. (Zanette et al., 2021). Consider positive scalars μ_1, μ_2 , and μ_3 , a parameter-dependent function $V : \mathbb{K} \times \Lambda_N \rightarrow \mathbb{R}_+$ satisfying

$$\mu_1 \|x\|^2 \leq V(x, \alpha) \leq \mu_2 \|x\|^2, \quad (8)$$

and a continuous and differentiable parameter-dependent function $\mathcal{V}_0 : \mathbb{K} \times \Lambda_N \times [0, T] \rightarrow \mathbb{R}$ satisfying

$$\mathcal{V}_0(x, \alpha, 0) = \mathcal{V}_0(x, \alpha, T), \quad (9)$$

for all $x \in \mathbb{K}$, $\alpha \in \Lambda_N$, and $T > 0$. Let $\mathcal{W}(x, \alpha, \tau) = V(x, \alpha) + \mathcal{V}_0(x, \alpha, \tau)$ and $\dot{\mathcal{W}} = \frac{d}{dt}[V(x, \alpha) + \mathcal{V}_0(x, \alpha, \tau)]$. If the inequality

$$\dot{\mathcal{W}}(x, \alpha, \tau) - \ell(x(t_k), \delta(t_k)) < -\mu_3 \|x\|^2 \quad (10)$$

holds, then:

- (1) $\Delta V(x(t_k), \alpha(t_k)) = V(x(t_{k+1}), \alpha(t_{k+1})) - V(x(t_k), \alpha(t_k)) < -\mu_3 T \|x(t_k)\|^2, \quad \forall k \in \mathbb{N}$,
- (2) The closed-loop system (7) is asymptotically stable under the event-triggering rule (5) with $T > 0$, $\forall(k, \alpha(t), \tau(t)) \in \mathbb{N} \times \Lambda_N \times [0, T]$.

As shown by Oliveira et al. (2025), the conservatism of the conditions can be reduced assuming $\mathcal{V}_0(x, \alpha, \tau)$ such that $\dot{\mathcal{W}}(x, \alpha, \tau)$ is a second-order polynomial in $\tau(t)$. To

this end, the following negativity lemmas for quadratic functions, based on Pólya's relaxation, will be used to derive LMI-based conditions.

Lemma 3. (Oliveira et al. (2025)). Let $\theta \in \Lambda_r$ and $f : \mathbb{R} \rightarrow \mathbb{R}^{p \times p}$, with $f(y) = a_2 y^2 + a_1 y + a_0$ a second-order polynomial matrix, where a_0, a_1 , and $a_2 \in \mathbb{R}^{p \times p}$. For a sufficient large $d \in \mathbb{N}$, $f(y) < \mathbf{0} \quad \forall y \in [\underline{h}, \bar{h}]$, if and only if all the matrix coefficients of the monomials of

$$(\theta_1^2 f(\bar{h}) + \theta_1 \theta_2 \mathcal{T}(\underline{h}, \bar{h}) + \theta_2^2 f(\underline{h})) (\theta_1 + \theta_2)^d, \quad (11)$$

with $\mathcal{T}(\underline{h}, \bar{h}) = (2\underline{h}\bar{h}a_2 + (\underline{h} + \bar{h})a_1 + 2a_0)$, are strictly negative-definite.

4. MAIN RESULTS

In this section, we propose conditions to design a controller (2) and an ETM (5) to address Problem 1. Assume the homogeneous polynomial matrix with $\theta_1, \theta_2 \in \Lambda_r$:

$$\Theta_d \equiv (\theta_1^2 \Pi_2 + \theta_1 \theta_2 \Pi_1 + \theta_2^2 \Pi_0) (\theta_1 + \theta_2)^d \quad (12)$$

where $d \in \mathbb{N}$, $\Pi_2 = T^2 \tilde{\Phi}_{2z} + T \tilde{\Phi}_{1z} + \tilde{\Phi}_{0z}$, $\Pi_1 = T \tilde{\Phi}_{1z} + 2 \tilde{\Phi}_{0z}$, $\Pi_0 = \tilde{\Phi}_{0z}$, $\tilde{\Phi}_{2z} = \begin{bmatrix} \tilde{\Phi}_2 & \mathbf{0}_{3n \times 2n} \\ \star & \mathbf{0}_{2n \times 2n} \end{bmatrix}$, $\tilde{\Phi}_{1z} = \begin{bmatrix} \tilde{\Phi}_1 & \mathbf{0}_{3n \times 2n} \\ \star & \mathbf{0}_{2n \times 2n} \end{bmatrix}$,

$$\tilde{\Phi}_{0z} = \begin{bmatrix} \tilde{\Phi}_0 & \tilde{M} & E_2^\top \tilde{X}^\top \\ \star & -\tilde{\mathcal{Q}}_\delta & \mathbf{0}_{n \times n} \\ \star & \star & -\tilde{\mathcal{Q}}_x \end{bmatrix}, \quad \tilde{\Phi}_2 = -\text{He} \left\{ \tilde{E}_1^\top \tilde{U} \tilde{E}_2 \right\},$$

$$\begin{aligned} \tilde{\Phi}_1 = & \text{He} \left\{ -E_3^\top \left(\tilde{Q}(\alpha(t)) E_2 + \tilde{S}(\alpha(t)) E_{12} \right) - E_{12}^\top \dot{\tilde{Q}}(\alpha(t)) E_2 \right. \\ & \left. + T \tilde{E}_1^\top \tilde{U} \tilde{E}_2 \right\} - 2 \tilde{E}_1^\top \tilde{U} \tilde{E}_1 - E_3^\top \tilde{R} E_3 - E_{12}^\top \dot{\tilde{S}}(\alpha(t)) E_{12} \\ & + \tilde{L}(\alpha(t)) \tilde{R}^{-1} \tilde{L}^\top(\alpha(t)), \\ \tilde{\Phi}_0 = & \text{He} \left\{ E_1^\top \tilde{P}(\alpha(t)) E_3 - E_{12}^\top \tilde{Q}(\alpha(t)) E_2 - \tilde{L}(\alpha(t)) E_{12} \right. \\ & \left. + (\epsilon E_1^\top + E_3^\top) (-A(\alpha(t)) \tilde{X} E_1 - B(\alpha(t)) \tilde{Z} E_2 + \tilde{X} E_3) \right. \\ & \left. + T \left(E_3^\top (\tilde{Q}(\alpha(t)) E_2 + \tilde{S}(\alpha(t)) E_{12}) + E_{12}^\top \dot{\tilde{Q}}(\alpha(t)) E_2 \right) \right\} \\ & + T \left(E_3^\top \tilde{R} E_3 + \tilde{E}_1^\top \tilde{U} \tilde{E}_1 + E_{12}^\top \dot{\tilde{S}}(\alpha(t)) E_{12} \right) - E_{12}^\top \tilde{S}(\alpha) E_{12} \\ & + E_1^\top \dot{\tilde{P}}(\alpha(t)) E_1, \end{aligned}$$

$\tilde{M} = -(\epsilon E_1^\top + E_3^\top) B(\alpha(t)) \tilde{Z} E_1$ with $E_1 = [\mathbf{I} \ \mathbf{0} \ \mathbf{0}]$, $E_2 = [\mathbf{0} \ \mathbf{I} \ \mathbf{0}]$, $E_3 = [\mathbf{0} \ \mathbf{0} \ \mathbf{I}]$, $E_{12} = E_1 - E_2$, $\tilde{E}_1 = [E_1^\top \ E_2^\top]^\top$, $\tilde{E}_2 = [E_3^\top \ \mathbf{0}]^\top$. Therefore, (12) is equivalent to (11).

Theorem 4. Consider the PETC system (1)-(4) and a sufficiently large $d \in \mathbb{N}$. Suppose there exist matrices $\mathbf{0} < \tilde{P}(\alpha(t)) \in \mathbb{S}^n$, $\tilde{S}(\alpha(t)) \in \mathbb{S}^n$, $\tilde{Q}(\alpha(t)) \in \mathbb{R}^{n \times n}$, $\tilde{L}(\alpha(t)) \in \mathbb{R}^{3n \times n}$, $\tilde{Z}(\alpha(t)) \in \mathbb{R}^{n_u \times n}$, $\mathbf{0} < \tilde{R} \in \mathbb{S}^n$, $\tilde{U} \in \mathbb{S}^{2n}$, $\tilde{X} \in \mathbb{R}^{n \times n}$, $\mathbf{0} < \tilde{\mathcal{Q}}_x \in \mathbb{S}^n$, $\mathbf{0} < \tilde{\mathcal{Q}}_\delta \in \mathbb{S}^n$, and a scalar $\epsilon > 0$, where all the matrix coefficients of the monomials of the homogeneous polynomial Θ_d (12) are strictly negative. Then the system (1) is asymptotically stable with the periodic event-triggered controller (2) whose robust gain matrix is given by $K = \tilde{Z} \tilde{X}^{-1}$ and the parameters of the ETM (5) given by $\mathcal{Q}_x = \tilde{\mathcal{Q}}_x^{-1}$ and $\mathcal{Q}_\delta = \tilde{X}^{-\top} \tilde{\mathcal{Q}}_\delta \tilde{X}^{-1}$.

Proof. Consider a parameter-dependent looped-functional, inspired by Seuret (2012), and given by $\mathcal{W}(x(t), \alpha(t), \tau(t)) = V(x(t), \alpha(t)) + \mathcal{V}_0(x(t), \alpha(t), \tau(t))$, where $V(x(t), \alpha(t)) = x^\top(t) P(\alpha(t)) x(t)$ and $\mathcal{V}_0(x(t), \alpha(t), \tau(t)) = (T - \tau) \left[(x(t) - x(t_k))^\top (S(\alpha(t)) (x(t) - x(t_k)) + 2Q(\alpha(t)) x(t_k)) \right] + \tau \zeta^\top U \zeta +$

$\int_{t-\tau(t)}^t \dot{x}^\top(s) R \dot{x}(s) ds$, with $\zeta = [x^\top(t) \ x^\top(t_k)]^\top$, positive definite matrices $P(\alpha(t)), R \in \mathbb{S}^n$, $U \in \mathbb{S}^{2n}$, $S(\alpha(t)) \in \mathbb{S}^n$, and $Q(\alpha(t)) \in \mathbb{R}^{n \times n}$, with a polytopic structure $[P(\alpha(t)) \ S(\alpha(t)) \ Q(\alpha(t))] = \sum_{i=1}^N \alpha_i(t) [P_i \ S_i \ Q_i]$, where $\alpha(t) \in \Lambda_N$. Note that $V(x(t), \alpha(t))$ and $\mathcal{V}_0(x(t), \alpha(t), \tau(t))$ satisfy (8) and (9). The time derivative of this looped-functional results in:

$\dot{\mathcal{W}}(x(t), \alpha(t), \tau(t)) = 2\dot{x}^\top(t)P(\alpha(t))x(t) + x^\top(t)\dot{P}(\alpha(t))x(t) - (x(t) - x(t_k))^\top S(\alpha(t))(x(t) - x(t_k)) + (T - \tau(t))2\dot{x}^\top(t)S(\alpha(t))(x(t) - x(t_k)) + (T - \tau(t))(x(t) - x(t_k))^\top \dot{S}(\alpha(t))(x(t) - x(t_k)) - 2(x(t) - x(t_k))^\top Q(\alpha(t))x(t_k) + (T - \tau(t))(2\dot{x}^\top(t)Q(\alpha(t))x(t_k) + 2(x(t) - x(t_k))^\top \dot{Q}(\alpha(t))x(t_k)) - \tau(t)\zeta^\top U \zeta + (T - \tau(t))\zeta^\top U \zeta + (T - \tau(t))\tau(t)2\dot{\zeta}^\top U \zeta + (T - \tau(t))\dot{x}^\top(t)R\dot{x}(t) - \int_{t-\tau(t)}^t \dot{x}^\top(s)R\dot{x}(s)ds$, with $\dot{\zeta} = [\dot{x}^\top(t) \ \mathbf{0}]^\top$. Next, considering $L(\alpha(t)) \in \mathbb{R}^{3n \times n}$ the last term of $\dot{\mathcal{W}}(x(t), \alpha(t), \tau(t))$ can be overbounded using the Jensen's inequality (Briat, 2011) by:

$$- \int_{t-\tau(t)}^t \dot{x}^\top(s)R\dot{x}(s)ds \leq -2\xi^\top(t)L(\alpha(t))(x(t) - x(t_k)) + \tau(t)\xi^\top(t)L(\alpha(t))R^{-1}L^\top(\alpha(t))\xi(t), \quad (13)$$

with $\xi = [x^\top(t) \ x^\top(t_k) \ \dot{x}^\top(t)]^\top$. Hence, from (5) and (10), we have that

$$\dot{\mathcal{W}} - \ell(x(t_k), \delta(t_k)) \leq \xi^\top(t)Q(\tau(t))\xi(t) - \delta^\top(t_k)Q_\delta\delta(t_k),$$

where $Q(\tau(t)) = \Phi_2\tau^2(t) + \Phi_1\tau(t) + \Phi_0 + E_2^\top Q_x E_2$, $\Phi_2 = -\text{He}\{\bar{E}_1^\top U \bar{E}_2\}$, $\Phi_1 = \text{He}\{-E_3^\top(Q(\alpha(t))E_2 + S(\alpha(t))E_{12}) - E_{12}^\top \dot{Q}(\alpha(t))E_2 + T\bar{E}_1^\top U \bar{E}_2\} - 2\bar{E}_1^\top U \bar{E}_1 - E_3^\top R E_3 - E_{12}^\top \dot{S}(\alpha(t))E_{12} + L(\alpha(t))R^{-1}L^\top(\alpha(t))$, $\Phi_0 = \text{He}\{E_1^\top P(\alpha(t))E_3 - E_{12}^\top Q(\alpha(t))E_2 - L(\alpha(t))E_{12} + TE_3^\top(Q(\alpha(t))E_2 + S(\alpha(t))E_{12}) + TE_{12}^\top \dot{Q}(\alpha(t))E_2\} + T(E_3^\top R E_3 + \bar{E}_1^\top U \bar{E}_1 + E_{12}^\top \dot{S}(\alpha(t))E_{12}) - E_{12}^\top S(\alpha(t))E_{12} + E_1^\top \dot{P}(\alpha(t))E_1$. From (7), we have $\mathcal{B} = [-A(\alpha(t)) \ -B(\alpha(t))K \ \mathbf{I} \ -B(\alpha(t))K]$, such that $\mathcal{B} \begin{bmatrix} \xi(t) \\ \delta(t_k) \end{bmatrix} = 0$. Assuming $\mathcal{X} = X^\top(\epsilon E_1^\top + E_3^\top)$, with $X \in \mathbb{R}^{n \times n}$, using the Finsler's Lemma (de Oliveira and Skelton, 2001), it follows that

$$\dot{\mathcal{W}} - \ell(x(t_k), \delta(t_k)) \leq \begin{bmatrix} \xi(t) \\ \delta(t_k) \end{bmatrix}^\top \Lambda(\tau(t)) \begin{bmatrix} \xi(t) \\ \delta(t_k) \end{bmatrix}, \quad (14)$$

with $\Lambda(\tau(t)) = \begin{bmatrix} \Phi_2\tau^2(t) + \Phi_1\tau(t) + \Phi_0 + E_2^\top Q_x E_2 & M \\ \star & -Q_\delta \end{bmatrix}$, $\Phi_0 = \hat{\Phi}_0 + \text{He}\{(\epsilon E_1^\top + E_3^\top)(-XA(\alpha(t))E_1 - XB(\alpha(t))KE_2 + XE_3)\}$ and $M = -(\epsilon E_1^\top + E_3^\top)XB(\alpha(t))KE_1$.

With $\tilde{X} = X^{-1}$, we multiply (14) by $\text{diag}\{\tilde{X}^\top, \tilde{X}^\top, \tilde{X}^\top, \tilde{X}^\top\}$ on the left and by its transpose on the right, and perform the following change of variables: $\tilde{P}(\alpha(t)) = \tilde{X}^\top P(\alpha(t))\tilde{X}$, $\tilde{S}(\alpha(t)) = \tilde{X}^\top S(\alpha(t))\tilde{X}$, $\tilde{Q}(\alpha(t)) = \tilde{X}^\top Q(\alpha(t))\tilde{X}$, $\tilde{U} = \text{diag}\{\tilde{X}^\top, \tilde{X}^\top\}U\tilde{X}$, $\tilde{R} = \tilde{X}^\top R \tilde{X}$, $\tilde{Q}_\delta = \tilde{X}^\top Q_\delta \tilde{X}$, $\tilde{L}(\alpha(t)) = \text{diag}\{\tilde{X}^\top, \tilde{X}^\top, \tilde{X}^\top\}L(\alpha(t))\tilde{X}$, and $\tilde{Z} = K\tilde{X}$. Finally, applying the Schur complement, (10) and (14) hold if

$$\begin{bmatrix} \tilde{\Phi}_2\tau^2(t) + \tilde{\Phi}_1\tau(t) + \tilde{\Phi}_0 & \tilde{M} & E_2^\top \tilde{X}^\top \\ \star & -\tilde{Q}_\delta & \mathbf{0}_{n \times n} \\ \star & \star & -\tilde{Q}_x \end{bmatrix} < \mathbf{0},$$

where $\tilde{Q}_x = Q_x^{-1}$, for $\tau(t) \in [0, T]$. Therefore, to ensure the negativity of the second-order polynomial, we use Lemma 3, resulting in (12). Note that block (3,3) of $\tilde{\Phi}_0$ implies $\tilde{X} + \tilde{X}^\top < 0$ and the non-singularity of $\tilde{X}$. $\square$

Remark 5. To verify the negativity of terms related to $\tilde{\Phi}_1$, as Π_2 and Π_1 , a Schur complement must be performed due to $\tilde{L}(\alpha(t))\tilde{R}^{-1}\tilde{L}^\top(\alpha(t))$.

Remark 6. Since all the coefficients of the monomials in (12) are strictly negative, Theorem 4 is solved through LMIs. In this paper, the parser ROLMIP (Agulhari et al., 2019) is used to get a finite set of LMIs from the parameter-dependent (and its derivatives) conditions. In case of unknown derivative bounds, impose all concerning matrices parameter-independent.

4.1 Optimization Procedures

To reduce the number of control updates as stated in Problem 1, we propose two optimization procedures to set Q_δ and Q_x for a given sampling period $T > 0$ and $\epsilon > 0$:

Convex optimization:

$$\begin{cases} \min \text{trace}(\tilde{Q}_x + \tilde{Q}_\delta) \\ \text{subject to: Theorem 4} \end{cases} \quad (15)$$

The optimization procedure (15) can indirectly reduce the number of transmissions in ETC schemes as shown by Zanette et al. (2021); de Souza et al. (2022); Peixoto et al. (2024). However, the minimization is not always effective, as shown by an example in the next section. To overcome this limitation, an algorithm using Differential Evolution optimization is proposed next.

Differential Evolution (DE) algorithm: The Differential Evolution (DE) algorithm is a global optimization method belonging to the family of evolutionary algorithms, biologically inspired by natural selection and evolution principles. Proposed by Storn and Price (1997), DE is particularly effective for optimization problems in continuous and high-dimensional search spaces. It operates on a population of parameter vectors, each representing a potential solution to the optimization problem. The key feature of DE is the *differential mutation*, which generates new candidate solutions by combining the differences between existing solutions in a population. Such a process enhances the algorithm's ability to explore the search space effectively. We use this optimization algorithm to search for matrices $\tilde{Q}_\delta$ and $\tilde{Q}_x$ of the ETM.

The DE works with a population, Z_t , $t = 1, \dots, N_G$, of N_p individuals, $z_{t,i}$ with $i = 1, \dots, N_p$, over a maximum of N_G generations aiming to minimize an objective function $\mathcal{O}(z)$ in a population of N . In our approach, the individual $z_{t,i}$ consists of $n(n+1)$ scalar numbers corresponding to the elements of the matrix pair $(\tilde{Q}_\delta, \tilde{Q}_x)$. Despite many developments in the DE algorithm, we use its simpler implementation (Storn and Price, 1997), where the optimization mechanism performs an initialization and then a loop on mutation, recombination, and selection, exploring the parameter space to minimize $\mathcal{O}(z)$. Within this

framework, the initialization generates a random initial population Z_1 with N_p individuals (potential solutions) verifying $z_\ell \leq z_{1,i} \leq z_h$, where z_ℓ and z_h are bounds chosen by the user. Next, in the loop, the mutation creates a population V_i of individuals $v_{t,i} = z_{t,b} + \eta(z_{t,r_1} - z_{t,r_2})$, with random indexes r_1 and $r_2 \in \{1, \dots, N_p\} \setminus \{i\}$ and η a scaling factor. The recombination combines individuals $z_{t,i}$ and $v_{t,i}$ to form a trial vector $u_{t,i}$ based on a crossover probability C_r and a random threshold. Finally, the selection consists of comparing $\mathcal{O}(z_{t,i})$ with $\mathcal{O}(v_{t,i})$, retaining the better-performing individual for the next generation. Each evaluation of $\mathcal{O}(z_{t,i})$ requires converting $z_{t,i}$ into the pair $(\tilde{Q}_\delta, \tilde{Q}_x)$ and then solving the convex optimization procedure given by Theorem 4 to get a stabilizing control gain K . If the LMIs are unfeasible, then $\mathcal{O}(z_{t,i}) = (1+g) \times 10^6$, where g is largest residue (eigenvalue) of the LMIs. Such an index is fundamental to conduct the DE to reach feasible individuals. Otherwise, if the LMIs are feasible, the trajectories of a set of 20 initial conditions are simulated and the mean control signal update economy (CSUE) is computed, which is assigned to $\mathcal{O}(z_{t,i})$, defined as

$$\text{CSUE} [\%] = \left(1 - \frac{\text{Number of control updates}}{\text{Number of periodic samples}}\right) 100\%.$$

5. NUMERICAL EXAMPLE

Consider the LPV system

$$\dot{x}(t) = \begin{bmatrix} -4.1 - 3\sigma(t) & 1 \\ 0 - 2\sigma(t) & 2 - 3.2\sigma(t) \end{bmatrix} x(t) + \begin{bmatrix} 3 \\ 2 \end{bmatrix} u(t), \quad (16)$$

drawn from (Zanette et al., 2021), with $|\sigma(t)| \leq 4$ and $|\dot{\sigma}(t)| \leq 5$. The objective is to evaluate the CSUE generated by the ETM obtained by the co-design procedure of Theorem 4 compared to the periodic sampling. Assuming a periodic sampling of $T = 0.015$ s, and applying the convex optimization procedure (15) for given values of $\epsilon > 0$ and Pólya's degree $d = 0, 1, 2, 3, 4$, we took 20 initial conditions in the extremity of a circle of radius 2 centered at the origin. For simulation purposes, $\sigma(t) = 4\sin(1.25t)$ for $0 < t < 2.2$ s and $\sigma(t) = 1.527$ for $t \geq 2.2$ s.

The CSUE for each value of ϵ and d is shown in the top plot of Fig. 2. In the bottom plot of Fig. 2, the objective function of the optimization procedure (15) is shown for different values of d . The range $8 \leq \epsilon \leq 16$ has been determined by a previous line search on ϵ values looking for a better ETC behavior. Note that even with the objective function decreasing as the degree d increases, the CSUE has an erratic behavior. This finding suggests that the objective function in (15) delivers some degree of CSUE, but it does not have a direct correlation with the main objective in Problem 1. However, it is clear that increasing Pólya's degree d relaxes the optimization procedure (15), leading to better values of the objective function.

This same challenge, the co-design of K in (2) and $\mathcal{Q}_\delta$ and $\mathcal{Q}_x$ in (5) to improve CSUE, has been solved by the DE Algorithm implemented using open-source Python packages `pymoo/pymode` with $N_p = 32$, $N_G = 500$, $C_r = 0.5$, and $0.5 \leq \eta \leq 1$ in 32 experiments. We repeated the simulations with $\epsilon = 10.8$ and took the same 20 initial conditions over a circle of radius 2 centered at the origin used in the previous approach. The percentage of the CSUE is compared between both and shown in the box plots in Fig. 3. Increasing the Pólya's degree d leads

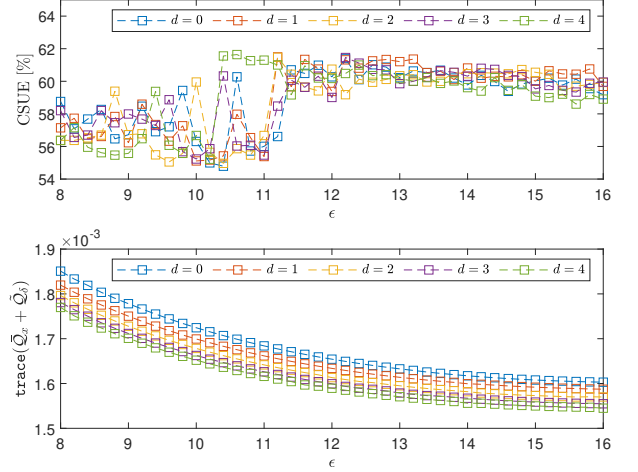


Fig. 2. Top plot: average CSUE related to 20 initial conditions for different values of ϵ using the convex optimization procedure (15); Bottom plot: objective function of the optimization for different values of ϵ .

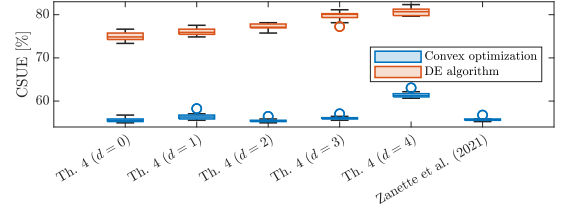


Fig. 3. Distribution of the percentage of the CSUE for 20 initial conditions when using the convex optimization (15) and the DE algorithm in Theorem 4. The DE algorithm can directly minimize the number of transmissions when increasing the degree d of Pólya's relaxation.

to a better economy when using the DE Algorithm. Also, note that both approaches can reach fewer transmissions than the emulation procedure of Zanette et al. (2021), which employs a parameter-dependent gain control law inherently less conservative. Remarkably, even with a fixed gain, independent of parameter variations, Theorem 4 demonstrates reduced network resource utilization.

Using the DE algorithm approach with Theorem 4 and $d = 4$, the co-design yields the robust control gain $K = [-3.1465 \ -16.0760]$ and the following triggering function parameters: $\mathcal{Q}_\delta = \begin{bmatrix} 11.9625 & 50.2691 \\ 50.2691 & 233.8616 \end{bmatrix}$, $\mathcal{Q}_x = \begin{bmatrix} 1.3425 & 0.6252 \\ 0.6252 & 0.2912 \end{bmatrix}$.

In Fig. 4, we present the simulation results for an initial condition $x(0) = [1 \ 0]^T$. The top plot displays the evolution of the states of the system $x(t)$, while the bottom plot shows the time between each triggering event (scaled on the left) and the control signal (scaled on the right). This simulation had a CSUE of 80.18% and a maximum triggering interval of 0.21 s, corresponding to 14 periodic sampling intervals ($T = 0.015$ s).

6. CONCLUSION

This work addressed the co-design of periodic event-triggered robust controllers for LPV systems. The selected parameter-dependent looped functionals led to second-

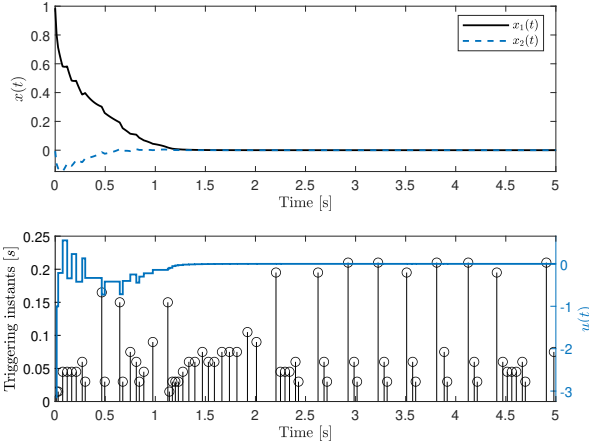


Fig. 4. Top plot: system's state, $x(t)$. Bottom plot: Triggering intervals and the control signal $u(t)$.

order polynomial stability conditions in the delay parameter, handled via Pólya's theorem to reduce conservatism. This has resulted in LMI-based conditions formulated to design the event-triggering mechanism and controller gain jointly. An iterative algorithm based on the differential evolution method has been employed to further reduce transmissions beyond standard convex optimization. A numerical example demonstrated improved efficiency in reducing control updates while ensuring stabilization using robust controllers. Future work will explore applications and more advanced event-triggering mechanisms for enhanced performance and LPV controllers.

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